Climate drivers of soft wheat yield losses in France

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This report links climate to soft wheat yield losses in the 13 metropolitan regions of France over the harvest years 2001 to 2019. After the data and the climate indices (Sections 1 to 3), it compares a Ridge baseline with a deep learning model trained on the daily climate of the season (Sections 4 to 7). Section 5 corrects three CMIP6 models and applies the baseline, which performs better, to 2031-2050 and 2071-2090 under SSP5-8.5. Finally Section 6 includes limitation and possible improvements.

1. Yield data and yield anomaly

Yields come from the Agreste SAA database, which reports yield, area and production for every French region from 2000 to 2020. The analysis uses crop code 1.11.1.11, which groups winter and spring soft wheat, and is restricted to the 13 metropolitan regions. Winter wheat covers on average 99% of the soft wheat area, and never less than 90% in a single region and year, so the series can safely be read as winter wheat. The series have no missing values, and the yield column agrees with production divided by area to within 0.005 t/ha. Besides the weather of the season, yields depend on factors that change slowly over the years, such as varieties and farming practices. This slow change, the trend, has to be removed first, so that what remains can be related to climate. A single national trend might seem enough, since the national yield, computed as total production over total area, is almost flat over the period (about +0.001 t/ha per year). The regional trends, however, differ even in sign. They are slightly negative in Bourgogne-Franche-Comté and Centre-Val de Loire and positive in Bretagne, Normandie and Hauts-de-France. Consequently, a single national trend would leave the regional trends inside the anomalies, and a region with a rising trend would show apparent losses in the early years and apparent gains in the late ones. The trend is therefore removed region by region. Following Ben-Ari et al. (2018) and Nóia Júnior et al. (2025), the yield anomaly is the relative deviation of the observed yield from a linear trend fitted separately for each region,

$$Y_{\mathrm{anm}}(i,t) = \frac{Y_{\mathrm{obs}}(i,t) - Y_{\mathrm{exp}}(i,t)}{Y_{\mathrm{exp}}(i,t)} \times 100, \qquad Y_{\mathrm{exp}}(i,t) = a_i + b_i\, t$$

where $Y_{obs}(i,t)$ is the yield observed in region i and harvest year t, $Y_{exp}(i,t)$ is the yield expected from the trend, and a_i and b_i are the intercept and slope of region i. An anomaly of −10% therefore means a yield 10% below what the trend of that region predicts for that year.

The anomaly is the target of all models, so the way the trend is fitted also matters for the evaluation. If the trend were fitted on all years, the yield of a test season would contribute to its own expected yield and would thus partly leak into its own target. Therefore, when the models are trained and evaluated, the trend of each region is fitted again on the training years only (`fit_trends` in `src/amigo_yield/baseline.py`), and the anomaly of the test season is computed from that trend. With trends fitted on all years, 31 of the 247 region-seasons show a loss larger than 10%, whereas with trends fitted on the training years 34 do. The evaluation uses the second definition.

Table 1 summarises the main loss seasons, which together account for 26 of the 31 losses. The 2016 loss is by far the largest and is concentrated in the centre and north, while 2003 hit mostly the centre, east and south, and 2007 mostly the west.

Table 1. Main loss seasons. National anomaly (area-weighted, trend on all years), number of regions with a loss larger than 10%, most affected regions, and climate of the season as monthly deviation from the regional normal, averaged over the 13 regions.

Season	National anomaly	Regions below −10%	Most affected regions	Climate signature
2016	−24.8%	9	Île-de-France −45%, Bourgogne-Franche-Comté −35%, Centre-Val de Loire −34%	Very mild late autumn (December 2015 +3.9 °C in maximum temperature), then a wet and cool May and June (253 mm in Île-de-France, the wettest of the period)
2003	−10.0%	8	Corse −37%, Auvergne-Rhône-Alpes −27%, Bourgogne-Franche-Comté −23%	Dry spring, then a hot June (+3.3 °C in maximum temperature)
2007	−9.9%	5	Bretagne −21%, Occitanie −14%, Pays de la Loire −14%	Mild winter, warm and dry April (+4.0 °C), then a wet June and July

Season	National anomaly	Regions below −10%	Most affected regions	Climate signature
2001	−5.1%	2	Corse −28%, Pays de la Loire −14%	Very wet autumn and winter (November 2000 +57 mm, January 2001 +41 mm)
2011	−4.7%	2	Nouvelle-Aquitaine −19%, Occitanie −13%	Warm and dry April and May (−42 and −56 mm)

2. Climate data

2.1 Data analysis

Climate is described by ERA5-Land daily maximum and minimum temperature, precipitation and the two wind components, on a 0.1° grid of 96 × 143 cells over France. Temperature and wind are available from 2000 to 2019 and precipitation from 1991 to 2022. Surface pressure is also available, but it is not used because the CMIP6 provided projections do not include it. The growing season runs from October of the year before harvest to July of the harvest year (Ben-Ari et al. ,2018), August and September are therefore excluded. Regional values are the mean of the grid cells whose centre falls inside the region polygon, with region codes that match those of Agreste. Three data issues had to be handled at this step. First, the files are not stored in the same way. Maximum temperature and precipitation store latitude from south to north and the other four variables from north to south, and precipitation stores its coordinates in double precision, so its latitudes differ from the others by about 0.000002 degrees. Since xarray matches grid cells by their coordinate values, the same cell would then appear as two different cells, and the regional mask, which is built on the grid of one variable, would not match the others. For this reason `read_era5` (`src/amigo_yield/climate.py`) rounds all coordinates to 0.01° and sorts latitude from south to north, so that every variable shares exactly the same grid. Second, a season is used only when every variable has data for all its ten months. Temperature and wind start in January 2000 and end in December 2019, so the season 2000 lacks them for October to December 1999 and the season 2020 lacks them for January to July 2020. Both seasons are therefore removed, which leaves the seasons 2001 to 2019, that is 247 region-seasons. Third, wind speed is computed from the daily mean components u and v. Winds that blow from opposite directions within the same day cancel out in these daily means, so the computed speed is lower than the true mean speed. Moreover, gusts last from seconds to minutes and disappear entirely in a daily mean. The wind variable therefore describes the large-scale circulation of the day and cannot represent the gusts that cause lodging. Finally, temperatures are converted from kelvin to degrees Celsius and precipitation from metres to millimetres of water. An exploratory correlation analysis (`notebook/02_climate_data.py`) uses the monthly deviations from the regional normal. Maximum and minimum temperature move together in every month (r = 0.78 to 0.96). Wind follows precipitation from October to April (r = 0.54 to 0.72), which is the typical signature of Atlantic depressions, and from March to July wetter months have lower maximum temperatures (r = −0.39 to −0.64). The monthly variables most strongly correlated with the yield anomaly of the season are November wind (r = −0.35), May minimum temperature (−0.34), December minimum and maximum temperature (−0.27 and −0.26), May maximum temperature (−0.25) and June precipitation (−0.23). These correlations are only indicative. Indeed, the 13 regions of a season share much of the same weather, so the 247 region-seasons are far from independent, and among 40 pairs of month and variable a few values around 0.3 can appear by chance. The November wind relation was examined further, since the crop is only emerging at that time and a direct effect of wind is unlikely. The correlation slightly changes after removing the part explained by precipitation (r = −0.30) or after excluding 2016 (r = −0.31), and a loss larger than 10% follows 26% of the Novembers windier than normal against 3% of the calmer ones. However, 20 of the 26 losses that follow a windy November belong to 2003, 2007 and 2016, whose losses have other documented causes. Windy Novembers therefore more plausibly mark stormy autumns with intense rainfall, waterlogged soils and delayed sowing, and the association rests on only three seasons. Over wider windows, moreover, wind follows precipitation from October to February (r = 0.65) and shows no relation with yield from May to July (r = 0.04). Wind is therefore not used by the models.

2.2 Climate indices

The climate indices summarise the conditions that the loss seasons of Table 1 and the literature associate with yield losses (Table 2). All seven indices and the interaction enter the final baseline.

Table 2. Climate indices of the baseline.

#	Index	Window	Definition	Reason	Source
1	precip_oct_jan	Oct to Jan	total precipitation (mm)	waterlogged soils at sowing and in winter	Zampieri et al. 2017; 2001 loss
2	tmean_nov_dec	Nov to Dec	mean of (tmax + tmin) / 2 (°C)	warm late autumn	Ben-Ari et al. 2018; 2016 loss
3	precip_mar_apr	Mar to Apr	total precipitation (mm)	spring drought; rain also favours foliar diseases	Nóia Júnior et al. 2025; 2003 and 2011 losses
4	precip_may_jun	May to Jun	total precipitation (mm)	wet spring around anthesis	Ben-Ari et al. 2018; 2016 loss
5	temperature_range_may_jun	May to Jun	mean of tmax − tmin (°C)	indirect signal of solar radiation	Hargreaves and Samani 1982
6	tmax_may_jun	May to Jun	mean of tmax (°C)	heat at anthesis and grain filling	Gammans et al. 2017; 2003 loss
7	hot_days_may_jul	May to Jul	days with tmax above 30 °C	heat stress at anthesis	Porter and Gawith 1999
+	interaction		index 2 × index 4, both standardised	compound event of 2016	Ben-Ari et al. 2018

The 30 °C threshold comes from experimental studies, which show that maximum temperatures above about 30 to 31 °C around anthesis reduce grain set (Porter and Gawith, 1999). These studies refer to the temperature of a site, whereas a regional mean smooths the extremes, so the threshold is exceeded on only 3.2% of the days between May and July and the hot day count is often zero. An index is set to missing when its window contains fewer than 28 days of data per month on average (for instance fewer than 112 days for October to January), so that no index is computed on incomplete data.

3. Baseline

3.1 Model and validation

The baseline is a Ridge regression on the standardised climate indices. With 19 seasons, a linear model with few inputs is the simplest model that can be checked in detail. Moreover, its coefficients can be interpreted, and its behaviour beyond the range of the training data is predictable. Each index is first standardised with region fixed effects (`standardise_indices` in `src/amigo_yield/baseline.py`). Specifically, the mean of the index in its region is subtracted, so that the model sees only how far a season lies from the normal of that region, and the result is divided by a single standard deviation computed over all regions. Provence, for example, is always warmer than Bretagne, and this permanent difference says nothing about the yield anomaly, which is already measured against the trend of each region. Since the standard deviation is common to all regions, one additional hot day also has the same weight everywhere. The interaction is the product of the two standardised indices, so it becomes large when both indices are far from normal in the same direction, as in 2016, when both were well above normal. All models are evaluated with a nested cross-validation grouped by year (Figure 2). In the outer loop, each of the 19 seasons is in turn the test year and is predicted by a model trained on the other 18, with all 13 regions of the test year held out together. Whole years are held out because the regions of a year share the same weather. In the inner loop, every choice on the model is made on the 18 training years only, again with validation sets made of whole years. For the Ridge regression the inner loop is itself a leave-one-year-out on the training years, which selects the penalty among seven values between 0.001 and 1000. For the network, it uses three blocks of six consecutive training seasons, described in Section 4. Consequently, the test year never enters the estimation of the yield trend, of the regional means and standard deviation of the inputs, or of any setting of the model.

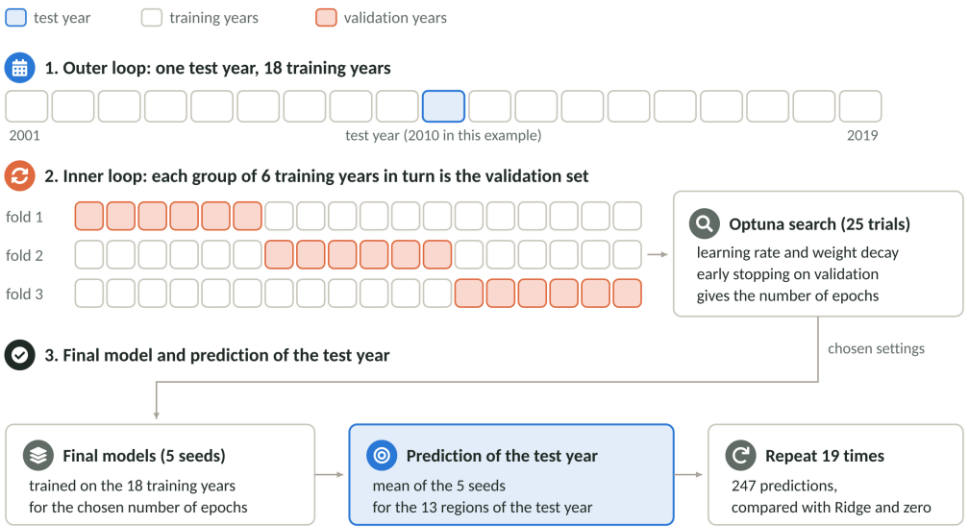


Figure 2. Nested cross-validation grouped by year used for the deep learning model, shown for one test year. The Ridge regression follows the same outer loop, with a leave-one-year-out as inner loop.

As a reference, a zero model always predicts a normal year, that is an anomaly of zero and a yield on the trend. All models are evaluated on the 247 out-of-sample predictions. The root mean squared error (RMSE) measures the typical error in points of anomaly, and the correlation between observed and predicted anomalies shows whether a model follows the direction of the changes from year to year. Since the target application is insurance, two further measures are reported. The loss hit rate is the share of losses larger than 10% that the model also predicts as losses, and the false alarm rate is the share of the other region-seasons that the model wrongly predicts as losses.

3.2. Experiments and results

The final configuration was reached through successive versions, and two indices and the standardisation were changed after a first validation run. Since choices made after seeing the results can make the final performance slightly optimistic, all versions are reported. A first version with five indices (all except the two heat indices), standardised within each region, reached an RMSE close to that of the zero model and predicted the 2003 loss with the wrong sign. Without a heat index, the model could not see the heatwave of June 2003, so the heat indices tmax_may_jun and hot_days_may_jul were added. The standardisation within each region then failed for the hot day count. In Provence-Alpes-Côte d'Azur, for instance, the regional mean exceeds 30 °C only in 2015 and 2019, so when these seasons are held out the count is zero in every training season, its standard deviation is zero and the standardisation divides by zero. The region fixed effects of Section 3.1 solve the problem, since their standard deviation is computed over all regions together, and they lowered the RMSE to 10.2 points of anomaly. The final baseline therefore uses the seven indices and the interaction of Table 2, for a total of nine parameters including the intercept. The Ridge regression explains part of the variability of seasons it has never seen (Table 3). It lowers the RMSE from 11.4 to 10.2 points of anomaly, recognises 7 of the 34 losses larger than 10% and raises a false alarm in 4.7% of the other region-seasons. The penalty selected inside the folds is 1 in 15 of the 19 folds.

Table 3. Performance on the 247 out-of-sample predictions

Model	RMSE (points of anomaly)	Correlation	Losses recognised	False alarms
Ridge on 5 indices + interaction standardised with each region	12.24	-0.10	0%	0%
Ridge on 7 indices + interaction standardised with region fixed effects	**10.2**	**0.46**	**21% (7 of 34)**	**4.7%**
Zero model (normal year)	11.4		0%	0%

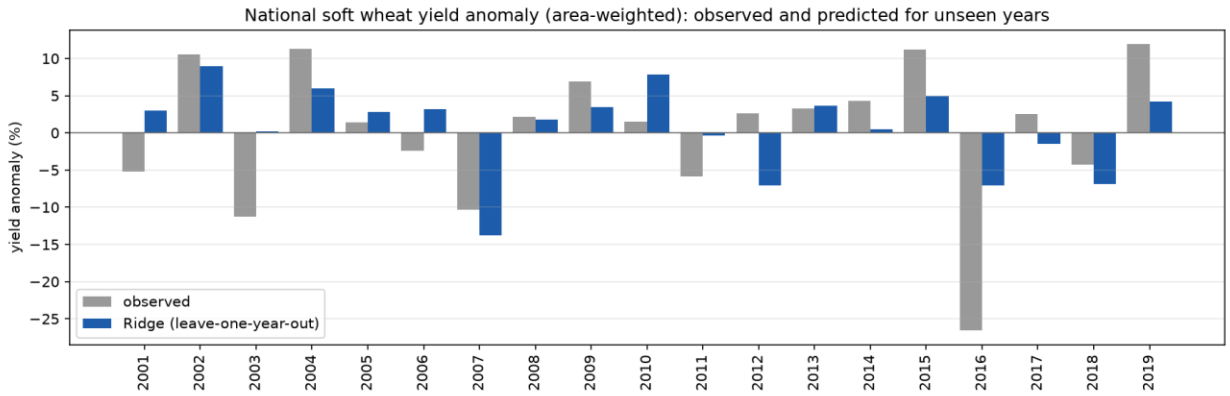


Figure 1. National yield anomaly (area-weighted), observed and predicted by the Ridge baseline for unseen seasons

At national level (Figure 1), the 2007 loss is predicted quite well (−14% against −10% observed). The 2016 loss is predicted in the right direction but at about a quarter of its size (−7% against −27%), since no training season contains a comparable compound event. The 2003 loss is missed (0% against −11%), and 2012 appears as a big false alarm (−7% against +3%).

Table 4. Coefficients of the Ridge model fitted on all seasons, in points of yield anomaly per standard deviation of the index above its regional mean. They serve for interpretation only; the performance comes from the cross-validation.

Index	Coefficient	Interpretation
tmax_may_jun	−7.6	a late spring hotter than normal lowers yields; the strongest effect
temperature_range_may_jun	+6.4	at equal maximum temperature, a wider daily range, that is more sunshine, raises yields
tmean_nov_dec × precip_may_jun	−3.7	a warm late autumn followed by a wet late spring adds a loss beyond the two separate effects
precip_oct_jan	−2.9	a wet autumn and winter lowers yields
precip_mar_apr	−1.6	a wet early spring lowers yields
tmean_nov_dec	−0.5	small effect on its own
hot_days_may_jul	+0.1	no effect beyond the mean maximum temperature
precip_may_jun	−0.1	no effect on its own; it acts through the interaction

These signs agree with the literature on French wheat. Late spring heat is the main driver of lower yields, in line with Gammans et al. (2017), who found that cereal yields in France respond negatively to warmer spring and summer temperatures. The positive effect of the temperature range reflects radiation, since clear days have a wider range (Hargreaves and Samani, 1982), and reduced solar radiation during flowering caused about 31% of the 2016 loss (Nóia Júnior et al., 2023). Since the two indices are correlated, however, their coefficients have to be read together. The interaction reproduces the compound event identified by Ben-Ari et al. (2018), a warm late autumn followed by a wet spring, whose two components add little when taken alone. The negative effect of rainfall from autumn to early spring agrees with Zampieri et al. (2017), who found French wheat more sensitive to water excess than to drought, and with the role of rain-borne diseases reported by Nóia Júnior et al. (2025). In particular, more rainfall in March and April lowers yields. Finally, the hot day count adds nothing beyond the mean maximum temperature on regional data.

4. Deep learning

4.1 Model and validation

The baseline only sees the windows chosen by hand. The deep learning model receives the daily climate of the whole season and learns by itself which periods matter, so the comparison between the two models shows whether the indices miss information contained in the daily series.

Each region-season is described by 304 days of maximum temperature, minimum temperature and precipitation, from 1 October to 31 July. The 29 February is removed, so that every position corresponds to the same calendar day in every season. Each variable is standardised with StandardScaler (https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html), that is with the mean and standard deviation of all the days of the training seasons, and the target is standardised in the same way and transformed back to

points of anomaly after prediction. Both scalers are fitted inside every fold, and the rows, the target and the outer folds are the same as for the baseline. The network is a one-dimensional convolutional network (`YieldCNN`). Each of its 2 filters spans 7 days and slides along the season, so it acts as a detector of a weekly weather pattern, for instance a hot and dry week, and gives a response for every day. A ReLU activation keeps only the positive responses, and a max pooling layer then splits the season into 10 blocks of about one month and keeps the strongest response of each block. Unlike an average, the maximum preserves short extremes such as a few very hot days. After a dropout of 0.2, a linear layer combines the 10 blocks of both filters, so the same pattern can weigh differently in autumn and in June. The network has 65 trainable parameters, namely 44 in the filters (2 filters × 3 variables × 7 days, plus 2 biases) and 21 in the linear layer (20 weights and 1 bias). This size was chosen for the inner training sets, which contain 12 of the 18 training seasons, about 156 region-seasons (Figure 2). Training uses the Adam optimiser with a mean squared error loss on batches of 32 seasons. In the inner loop (Figure 2), Optuna (TPE sampler with a fixed seed, 25 trials) searches the learning rate between 0.0001 and 0.01 and the weight decay between 0.00001 and 0.1, both on a logarithmic scale. Every trial is trained once per inner fold with early stopping, which ends the training when the validation error has not improved for 20 epochs, up to 300, and keeps the weights of the best epoch. The score of a trial is the mean of its best validation errors, and its number of epochs is the mean of its best epochs. The final network of each outer fold is then trained on all 18 training seasons with the chosen settings, five times with different seeds, which change the initial weights, the order of the batches and the dropout masks. The five predictions are averaged, so that the result does not depend on a single initialisation (Paudel et al., 2023). The number of filters, the filter length and the dropout rate are fixed. Monte Carlo Dropout (Gal and Ghahramani, 2016) was selected as a measure of the model uncertainty, but it is switched off in the final run because model performance was poor. In addition, with 100 Monte Carlo Dropout passes its 80% interval contained only 26% of the observed anomalies, whereas a well calibrated interval should contain about 80% of them. The intervals are therefore far too narrow, mainly because Monte Carlo Dropout describes only the uncertainty of the network weights, while most of the error comes from factors that regional climate does not describe.

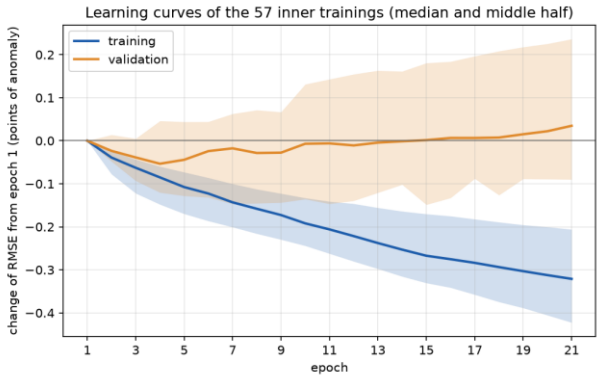


Figure 3. Learning curves of the 57 inner trainings with the chosen settings, as change of RMSE from the first epoch. Lines: median; bands: middle half.

The training and validation errors are recorded after every epoch (Figure 3). Over the 57 inner trainings (19 outer folds × 3 inner folds), the median training error falls by 0.32 points of anomaly in the first 21 epochs, while the median validation error rises by 0.03. In other words, the network learns the training seasons without improving on unseen ones, and early stopping accordingly selects between 3 and 55 epochs, with a median of 16.

4.2 Experiments and results

The experiments reported here start from the 2-filter network. The 2-filter network was first trained on inputs expressed as deviations from daily regional normals. Specifically, each day of the season was compared with the mean of the same calendar day in the same region over the training seasons, and the difference was divided by one standard deviation per variable. Each daily normal, however, is a mean of only about 18 values and is therefore noisy, so this normalisation adds noise to every input day. It was therefore replaced by StandardScaler, which uses a single mean and standard deviation per variable. The RMSE fell from 11.65 to 11.38 points of anomaly and the correlation rose from −0.16 to 0.14. Since this change was made after looking at the outer results, the comparison between the two configurations is slightly optimistic.

Table 5 compares the three models on the same 247 out-of-sample predictions. The final network reaches the level of the zero model without going below it, and it recognises none of the 34 losses. Its predictions vary very little around zero, so it misses both 2016 and 2003, whereas the Ridge regression predicts the direction of the 2016 loss and spreads its

predictions about six times more. With 19 seasons, the network cannot learn from the daily series which periods of the season matter, while the indices bring that knowledge from the literature. Moreover, the network has seven times more parameters and takes about ten times longer to evaluate. The projections of Section 5 therefore use the Ridge regression.

Table 5. Comparison of the three models on the 247 out-of-sample predictions (results/baseline and results/dl: metrics.csv, yearly.csv and config.json).

	Ridge on indices	1D CNN on daily climate	Zero model
Inputs	7 indices and their interaction	304 days × 3 variables	none
Trainable parameters	9	65	0
RMSE (points of anomaly)	10.22	11.38	11.43
Correlation	0.46	0.14	
Losses larger than 10% recognised	7 of 34 (21%)	0 of 34	0 of 34
False alarms	4.7%	0%	0%
Standard deviation of the predictions (observed 11.5)	6.3	1.1	0
National anomaly 2016 (observed −26.6%)	−7.1%	+0.3%	0%
National anomaly 2003 (observed −11.3%)	+0.2%	+0.1%	0%
Training time	48 s	8.6 min	
Peak memory	0.44 GB	0.55 GB	

Training times and memory in Table 5 refer to a laptop with an Intel Core i7-8750H processor and 16 GB of RAM, using one CPU thread, and cover the whole nested cross-validation. The whole pipeline runs in 35 minutes, most of which is spent on the bias correction of Part C.

5. Loss-driving conditions under climate change

The projections come from three models (CNRM-ESM2-1, EC-Earth3 and IPSL-CM6A-LR), with daily maximum and minimum temperature and precipitation for the historical experiment (1985-2014) and the SSP5-8.5 scenario (2015-2090). According to the file metadata, the data are on the grids distributed by the modelling centres, with no bias adjustment. These grids have 8 × 12, 18 × 25 and 10 × 7 cells over France, so every model is interpolated bilinearly to the ERA5-Land grid, masked with the ERA5-Land land cells and averaged over the regions with the same mask as ERA5-Land. Temperature is converted from kelvin and precipitation from kg m^{-2} s^{-1} to mm per day.

5.1 Comparison with ERA5-Land and bias correction

The models are compared with ERA5-Land over 2000-2014, the overlap between the historical experiment and the reanalysis. Since model years do not correspond to observed years, the comparison uses the distributions of the daily regional values from October to July. EC-Earth3 is the closest to ERA5-Land, CNRM-ESM2-1 is too warm, and IPSL-CM6A-LR has cool maximum and warm minimum temperatures (Table 6). All three models are too wet, with too many days of light rain, and the biases grow in the tails. They also change with the month, since minimum temperature is too warm mostly from December to February and precipitation is overestimated from November to April (Figure 4, top).

Table 6. Difference between the raw models and ERA5-Land, daily regional values, October to July 2000-2014.

Model	tmax mean (°C)	tmax p99 (°C)	tmin mean (°C)	tmin p99 (°C)	precip mean (%)	precip median (mm/day)
CNRM-ESM2-1	+1.2	+2.7	+1.5	+4.3	+18	+0.34
EC-Earth3	+0.1	+1.5	+0.3	+1.3	+5	+0.20
IPSL-CM6A-LR	−0.8	+0.9	+1.7	+3.6	+23	+0.57

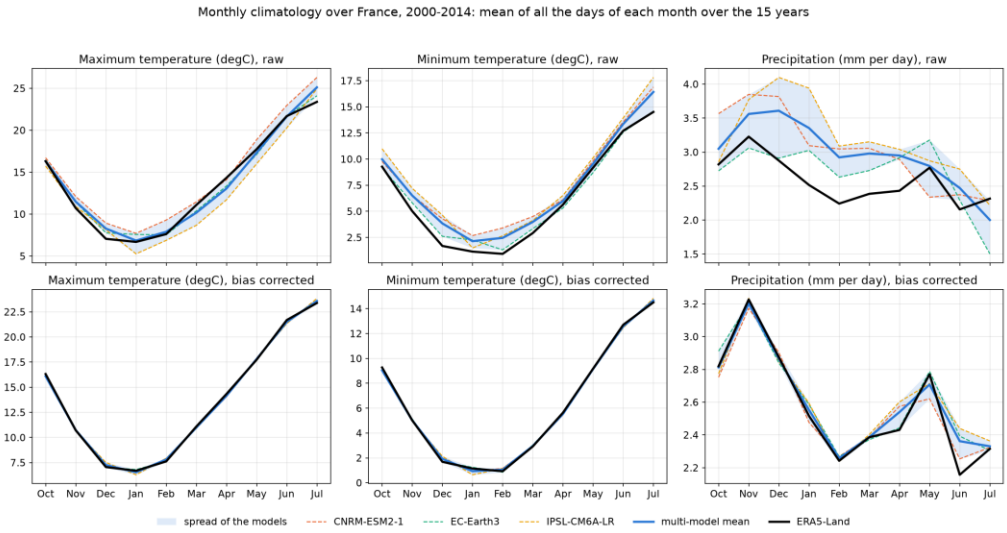


Figure 4. Monthly climatology over France, 2000-2014, of ERA5-Land and of the three models before (top) and after (bottom) the bias correction. The band spans the lowest and highest model.

Since the biases depend on the quantile and on the month, a correction of the mean alone would leave them in the tails, precisely where the loss-driving conditions lie. The models are therefore corrected with quantile delta mapping (Cannon et al., 2015), as implemented in the xsdba package (`train_bias_correction` and `apply_bias_correction` in `src/amigo_yield/cmip6.py`). In simple terms, the method compares the quantiles of the model with those of ERA5-Land over the calibration period, separately for every grid cell and calendar month with 50 quantiles. For each future day it then computes the change that the model simulates for that quantile, corrects the quantile for its calibration bias and adds the change back, so the bias is removed while the climate change signal of the model is kept. The correction is additive for temperature and multiplicative for precipitation. Dry days need a further step. Many days have exactly zero precipitation and cannot be ranked into quantiles, so values below 0.05 mm per day are replaced by small random values before the correction, and corrected values below the same threshold are set back to zero, following the reference implementation of the method. The correction is calibrated on 2000-2014 only. This period was not extended with the first SSP5-8.5 years, since the bias-corrected CMIP6 products that were checked, NASA NEX-GDDP-CMIP6 (Thrasher et al., 2022) and GDPCIR (Gergel et al., 2024), calibrate on the historical experiment only. After the correction, the mean of every model and variable lies within 0.05 °C or 0.05 mm per day of ERA5-Land, and the 99th percentile of temperature within 0.35 °C (Figure 4, bottom). Since this check uses the calibration years, it confirms the implementation without measuring the skill of the correction on independent years. Each future window of 20 harvest seasons, from October 2030 to July 2050 and from October 2070 to July 2090, is corrected on its own, so that the method keeps the change simulated by each model. Indeed, the warming of the seasonal mean of maximum and minimum temperature is the same before and after the correction to within 0.1 °C. The results are computed for every model separately and then averaged (ensemble mean), and the lowest and highest model give the spread. A second way of forming the ensemble, averaging the daily values of the three models before computing the indices (multi-model mean), was also tested. Since the days of different models do not correspond, their average flattens the extremes. In 2031-2050, for instance, the 99th percentile of maximum temperature rises by 1.1 to 2.8 °C in the three models, whereas in the multi-model mean it falls by 0.5 °C. The multi-model mean is therefore shown only for comparison.

5.2 Frequency of the loss-driving conditions

Relative to ERA5-Land 2001-2019, the mean maximum temperature of the season rises by 0.4 to 1.5 °C in 2031-2050 and by 2.4 to 4.5 °C in 2071-2090, and the hottest days warm more (99th percentile +5.3 to +7.2 °C by 2071-2090). Mean precipitation changes by less than 5%, but it falls on fewer days, with a lower median and a higher 99th percentile. Table 7 reports the mean of each index and the share of future seasons beyond the extremes of the present. These extremes are the 10th and 90th percentiles of each region in 2001-2019, so by construction about 10% of the present seasons lie beyond each of them.

Table 7. Climate indices, mean over regions and seasons. Ensemble mean of the corrected models, with the lowest and highest model in brackets. Last column: share of the 2071-2090 seasons beyond the present extreme of the region

Index	Present 2001-2019	2031-2050	2071-2090	Beyond present extreme, 2071-2090
tmax_may_jun (°C)	19.8	21.1 (20.1-21.9)	24.0 (22.5-25.2)	91% above p90 (76-99)
hot_days_may_jul (days)	3.0	7.9 (5.6-9.7)	23.3 (17.5-26.6)	88% above p90 (80-93)
tmean_nov_dec (°C)	6.3	7.0 (6.6-7.5)	9.4 (8.4-10.1)	86% above p90 (69-96)
temperature_range_may_jun (°C)	8.8	9.0 (8.7-9.3)	9.5 (9.0-10.0)	49% above p90 (27-72)
precip_may_jun (mm)	155	148 (131-161)	125 (108-153)	41% below p10 (22-53)
precip_oct_jan (mm)	345	356 (346-373)	373 (355-387)	25% above p90 (21-28)
precip_mar_apr (mm)	145	149 (140-160)	141 (137-146)	17% below p10 (13-21)

Late spring heat is the clearest signal. A May to June as hot as one season in ten today occurs in half of the seasons by 2031-2050 (26 to 64% across models) and in nine seasons in ten by 2071-2090. Late autumns become warmer and late springs drier, while wet late springs keep about their present frequency (11%, 5 to 18% across models). The three models agree on the warming of late autumn and late spring in both windows, and they differ mostly on precipitation.

5.3 Yield losses predicted by the baseline

The Ridge regression of the baseline, fitted on all 247 region-seasons with the indices and penalty of Section 6, is applied to the indices of every corrected model. The future indices are standardised with the regional means and the standard deviation of 2001-2019, so the model sees how far each future season lies from the present normal. This use is only partly legitimate, for two reasons. First, the model learns the variability from year to year of 2001-2019 and assumes the technology of that period, with no adaptation of varieties or sowing dates and no effect of higher CO_2, so its predictions describe the effect of climate alone. Second, at least one index lies outside the 2001-2019 range in 16 to 48% of the region-seasons of 2031-2050 and in 66 to 95% of those of 2071-2090, mostly late spring temperature and hot days. There, a linear model simply extends the slope estimated on milder seasons.

Table 8. Region-seasons with a predicted loss larger than 10% and mean predicted anomaly, SSP5-8.5. Ensemble mean of the three corrected models, with the lowest and highest model in brackets.

	Present 2001-2019	2031-2050	2071-2090
Seasons with a predicted loss, ensemble mean	7%	28% (16-45)	58% (52-65)
Seasons with a predicted loss, multi-model mean		16%	70%
Mean predicted anomaly, ensemble mean	0.0%	−4.6% (−9.2 to −1.8)	−13.9% (−16.8 to −10.6)
Observed loss seasons	13%		

With these caveats, the share of seasons with a predicted loss larger than 10% rises from 7% today to 28% in 2031-2050 and to 58% in 2071-2090 (Table 8, Figure 5). The Ridge regression already underestimates losses in the present, with 7% predicted against 13% observed, so the change between periods is more reliable than the levels. By mid-century the result depends strongly on the climate model, whereas by 2071-2090 the three models agree that a loss season becomes the most common outcome. The multi-model mean gives 16% and 70%, which shows how much the result depends on the way the ensemble is formed. The late-century value therefore indicates direction and order of magnitude, while the index frequencies of Table 7, which require no extrapolation, are the more robust result.

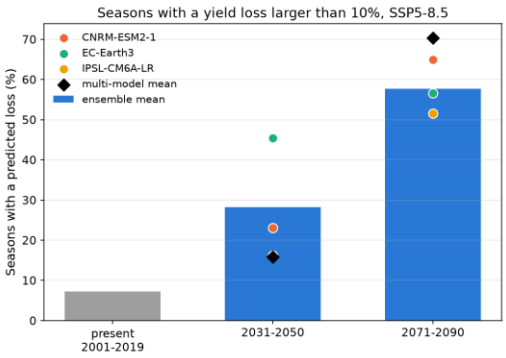


Figure 5. Share of region-seasons with a predicted yield loss larger than 10%, SSP5-8.5 (`results/cmip6/loss_seasons.png`). Bars: ensemble mean of the three corrected models; dots: single models; diamond: multi-model mean.

6. Limitations and possible improvements

The main limitation is the number of seasons. With 19 seasons and only three national losses close to or beyond 10%, an event without precedent in the training years, such as 2016, cannot be learned from the others. Important drivers are missing, since no data on solar radiation, soil moisture or crop diseases are used, and regional means include mountains and forests and smooth the extremes. Longer series, with Agreste data before 2000 and ERA5 data from 1950, would add past extreme seasons comparable to 2016, and radiation, soil moisture and satellite vegetation indices would cover the missing drivers. Finally, a larger computational budget would allow gridded inputs, a wider hyperparameter search and probabilistic outputs, so that every forecast comes with a calibrated range useful for pricing risk. For the deep learning model, four further steps would be worth testing. A loss that weights underestimated losses more than other errors would push the network away from predictions close to zero, which is what matters for an insurer. An explainability method would show which days and variables drive each prediction, and would check whether the network focuses on May and June as the literature suggests. Daily inputs aligned on crop stages through accumulated growing degree days, together with the indices of the baseline, would give the network the knowledge that 19 seasons cannot provide. Finally, pretraining the network on a task that needs no yield data, such as predicting the climate indices of the season from ERA5 since 1950, would let the network learn the structure of the daily weather before fine-tuning it on the 19 labelled seasons.

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